

Linear Operators: Basic Theory

Pathak – Nonlinear Analysis and Fixed Point Theory (2018)

Lecture Slides

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Chapter 5d: Bounded Linear Operators and Operator Spaces

Outline

- 1 Introduction to Linear Operators
- 2 Bounded Linear Operators
- 3 Spaces of Bounded Linear Operators
- 4 Convergence in Operator Spaces
- 5 Key Theorems and Applications
- 6 Computational Examples
- 7 Summary and Applications

What is an Operator?

Definition

An **operator** $A : X \rightarrow Y$ maps vectors from one space X (domain) to another space Y (codomain).

Key Concept

An operator is a generalization of the concept of a function to vector spaces. It can be thought of as an object that “acts on” vectors to transform them.

Domains of application:

- Matrix equations: $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$
- Integral transforms: $A : L^2[a, b] \rightarrow L^2[a, b]$
- Differential equations: $A : C[a, b] \rightarrow C[a, b]$
- Functional analysis: $A : X \rightarrow Y$ for Banach spaces X, Y

Linear Operators

Definition

Let X and Y be vector spaces over the same field $\mathbb{K}$ (where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$), and let $A : X \rightarrow Y$ be an operator with domain $D(A)$ and range $R(A)$. Then A is a **linear operator** if:

- ① **Additivity:** $A(x + y) = Ax + Ay$ for all $x, y \in X$
- ② **Homogeneity:** $A(\alpha x) = \alpha A(x)$ for all $x \in X, \alpha \in \mathbb{K}$

Equivalent Formulation

A is linear if and only if

$$A(\alpha x + \beta y) = \alpha A(x) + \beta A(y)$$

for all $x, y \in X$ and $\alpha, \beta \in \mathbb{K}$.

If $Y = \mathbb{R}$ (or $\mathbb{C}$), then A is called a **linear functional**.

Examples of Linear Operators

Example (Dot Product on $\mathbb{R}^n$)

Let $X = \mathbb{R}^n$, $Y = \mathbb{R}$, and $A : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$Ax = \sum_{i=1}^n a_i x_i \quad (\text{linear functional})$$

Example (Shift Operator on ℓ^2)

Let $X = Y = \ell^2$ (square-summable sequences). Define:

$$Ax = \left(x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_n}{n}, \dots \right)$$

Check: $A(\alpha x + \beta y) = \alpha Ax + \beta Ay$ ✓

Example (Integration on $L^2[0, 2\pi]$)

Let $X = L^2[0, 2\pi]$, $Y = \mathbb{R}$, and

$\int_0^{2\pi}$

Properties of Linear Operators

Theorem (Proposition 1.7)

Let X and Y be linear spaces and $A : X \rightarrow Y$ a linear operator. Then:

- ① $A(0) = 0$ (maps zero element to zero)
- ② The range $R(A) = \{y \in Y : y = Ax \text{ for some } x \in X\}$ is a linear subspace of Y
- ③ A is one-to-one (injective) if and only if $Ax = 0 \implies x = 0$
- ④ If $A : X \rightarrow Y$ is one-to-one, then A^{-1} exists on $R(A)$ and is linear
- ⑤ If $\dim(X) = n < \infty$ and A^{-1} exists, then $\dim(R(A)) = n$

Continuity and Linearity

Theorem (Theorem 1.12)

Let X and Y be normed spaces and $A : X \rightarrow Y$ a linear operator. If A is continuous at a single point in X , then A is continuous throughout space X .

Proof Sketch.

Suppose A is continuous at x_0 and $x_n \rightarrow x$. Then:

$$\begin{aligned}\|Ax_n - Ax\| &= \|A(x_n - x + x_0) - Ax_0\| \\ &\rightarrow 0 \quad (\text{as } x_n \rightarrow x)\end{aligned}$$

Thus A is continuous at every point $x \in X$. □

Definition of Bounded Linear Operator

Definition

Let X and Y be normed spaces and $A : X \rightarrow Y$ a linear operator. Then A is **bounded** if there exists a constant $M > 0$ such that

$$\|Ax\| \leq M\|x\| \quad \text{for all } x \in X.$$

Important Distinction

A bounded operator does NOT map the entire space X into a bounded set in Y . Rather:

- A bounded operator maps **bounded sets** to **bounded sets**
- The converse is also true: bounded sets $\rightarrow$ bounded sets $\Rightarrow$ bounded operator

Linear Functional

A linear functional $f : X \rightarrow \mathbb{R}$ is bounded if

$$|f(x)| \leq M\|x\| \quad \text{for all } x \in X.$$

Example: Unbounded Linear Operator

Example (Unbounded Functional on c_{00})

Let c_{00} be the space of finitely nonzero real sequences with supremum norm $\|\cdot\|_\infty$. Define $A : c_{00} \rightarrow \mathbb{R}$ by:

$$Ax = \sum_{i=1}^n i \cdot x_i$$

for $x = (x_1, x_2, \dots, x_n, 0, 0, \dots)$.

Why is it unbounded?

- Let $x_n = (0, 0, \dots, 0, 1/n, 0, \dots)$ with the 1 in position n
- Then $\|x_n\|_\infty = 1/n \rightarrow 0$
- But $A(x_n) = n \cdot (1/n) = 1 \not\rightarrow 0$
- So the operator is **discontinuous** at 0, hence unbounded

Bounded $\Leftrightarrow$ Continuous for Linear Operators

Theorem (Theorem 1.13 – Central Result)

Let X and Y be normed spaces and $A : X \rightarrow Y$ a linear operator. Then A is bounded if and only if it is continuous.

Bounded $\Rightarrow$ Continuous

- If $\|Ax\| \leq M\|x\|$ for all x
- Then $x_n \rightarrow 0 \implies$
- $\|Ax_n\| \leq M\|x_n\| \rightarrow 0$
- So A is continuous at 0
- By Thm 1.12: continuous everywhere

Continuous $\Rightarrow$ Bounded

- Proof by contrapositive
- If unbounded: $\|Ax_n\| > n\|x_n\|$ for some sequence
- Set $y_n = x_n/(n\|x_n\|)$
- Then $y_n \rightarrow 0$ but $\|Ay_n\| > 1$
- So A is discontinuous at 0

Key Insight

For linear operators, boundedness and continuity are equivalent! This is a **fundamental theorem** in functional analysis

Operator Norm

Definition

For $A \in B(X, Y)$ (space of bounded linear operators), the **operator norm** is:

$$\|A\|_B = \inf\{M : \|Ax\| \leq M\|x\|, \forall x \in X\}$$

Equivalent Formulations

All of the following are equivalent:

$$\begin{aligned}\|A\|_B &= \sup\left\{\frac{\|Ax\|}{\|x\|} : x \neq 0, x \in X\right\} \\ &= \sup\{\|Ax\| : \|x\| \leq 1, x \in X\} \\ &= \sup\{\|Ax\| : \|x\| = 1, x \in X\}\end{aligned}$$

Example (Matrix Operator Norm)

For $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (a matrix), $\|A\|_B$ equals the largest singular value of A .

Properties of the Operator Norm

Theorem

The operator norm on $B(X, Y)$ satisfies:

- ① **Norm property:** $\|Ax\| \leq \|A\|_B \cdot \|x\|$ for all $x \in X$
- ② **Triangle inequality:** $\|A + B\|_B \leq \|A\|_B + \|B\|_B$
- ③ **Homogeneity:** $\|\lambda A\|_B = |\lambda| \cdot \|A\|_B$ for scalar λ
- ④ **Composition:** $\|A \circ B\|_B \leq \|A\|_B \cdot \|B\|_B$
- ⑤ **Identity operator:** $\|I\|_B = 1$ where I is the identity

Banach Space Structure

With the operator norm, $B(X, Y)$ becomes a normed space. It is a **Banach space** if Y is a Banach space.

The Space $B(X, Y)$

Definition

Let X and Y be normed spaces. The space $B(X, Y)$ consists of all bounded linear operators from X into Y .

$B(X, Y)$ is a **linear space** with operations:

- **Addition:** $(A_1 + A_2)(x) = A_1x + A_2x$
- **Scalar multiplication:** $(\lambda A)(x) = \lambda A(x)$

Banach Space Property

If Y is a Banach space, then $B(X, Y)$ is also a **Banach space** with the operator norm.

Examples

- $B(\mathbb{R}^n, \mathbb{R}^m)$: all $m \times n$ real matrices
- $B(C[a, b], C[a, b])$: multiplication by continuous functions bounded by their maximum
- $B(L^2(\Omega), L^2(\Omega))$: integral operators with finite norm

Uniform Boundedness Principle

Theorem (Theorem 1.14)

Let X be a Banach space, Y a normed space, and $\{A_i\}_{i \in I}$ a family of bounded linear operators from X into Y such that $\{A_i x : i \in I\}$ is bounded for each $x \in X$.

Then $\{A_i\}_{i \in I}$ is a bounded family, i.e., the operators are **uniformly bounded**:

$$\sup_{i \in I} \|A_i\|_B < \infty$$

Intuition

If a family of operators is *pointwise bounded* (bounded at every point), then it is *uniformly bounded* (bounded with a single constant for all operators).

Importance

This is a **fundamental principle** in functional analysis with crucial applications to convergence of operator sequences and iterative methods.

Pointwise Limit of Operator Sequences

Theorem (Theorem 1.15)

Let X and Y be Banach spaces and $\{A_n\}$ a sequence of bounded operators from X into Y . If for each $x \in X$, the sequence $\{A_n x\}$ converges to Ax in Y , then:

- 1 A is a bounded linear operator, i.e., $A \in B(X, Y)$
- 2 $\|A\|_B \leq \liminf_{n \rightarrow \infty} \|A_n\|_B$

Proof Outline.

- 1 Linearity of A follows from linearity of A_n and continuity of limits
- 2 Boundedness of $\{A_n x\}$ for each x combined with Uniform Boundedness Principle gives uniform boundedness
- 3 Thus $\|A_n\|_B \leq M$ for all n , proving A is bounded



Types of Convergence for Operator Sequences

Definition

Let $\{T_n\}$ be a sequence in $B(X, Y)$ where $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are normed spaces:

- 1 **Uniform convergence:** $\|T_n - T\|_B \rightarrow 0$ as $n \rightarrow \infty$
- 2 **Strong convergence:** $\|T_n x - T x\|_Y \rightarrow 0$ as $n \rightarrow \infty$ for all $x \in X$
- 3 **Weak convergence:** $|f(T_n x) - f(T x)| \rightarrow 0$ as $n \rightarrow \infty$ for all $x \in X$ and $f \in Y^*$
(where Y^* is the dual space of bounded linear functionals on Y)

Hierarchy of Convergence

Theorem

The following implications hold for operator sequences in $B(X, Y)$:

Uniform Convergence $\Rightarrow$ Strong Convergence $\Rightarrow$ Weak Convergence

Uniform $\Rightarrow$ Strong If $\|T_n - T\|_B \rightarrow 0$, then:

$$\|T_n x - T x\| \leq \|T_n - T\|_B \|x\| \rightarrow 0 \forall x$$

Strong $\Rightarrow$ Weak If $T_n x \rightarrow T x$, then:

$$|f(T_n x - T x)| \leq \|f\| \|T_n x - T x\| \rightarrow 0 \forall x, f$$

No reverse implication Weak convergence does NOT imply strong, and strong does NOT imply uniform.

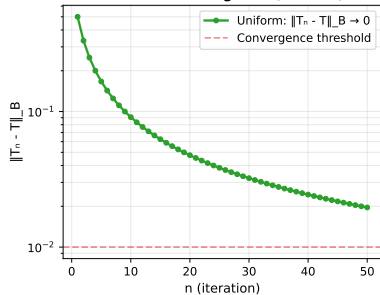
These are proper inclusions.

Convergence Speed

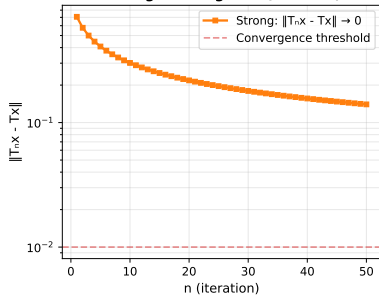
Uniform is fastest, weak is slowest. The choice depends on the application and available regularity.

Convergence Visualization

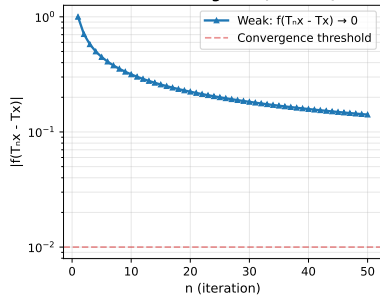
Uniform Convergence (Fastest)



Strong Convergence (Medium)



Weak Convergence (Slowest)



Corollary: Weakly Bounded Sets are Bounded

Corollary (Corollary 1.1)

Let C be a nonempty subset of a Banach space X . For each $f \in X^*$, suppose $f(C) = \{(x, f) : x \in C\}$ is a bounded set in $\mathbb{R}$. Then C is bounded.

Proof Idea

- 1 For each $x \in C$, define $T_x(f) = (x, f)$
- 2 Each T_x is a bounded linear operator: $T_x \in B(X^*, \mathbb{R})$
- 3 Boundedness of $f(C)$ means $\sup_{x \in C} |T_x(f)| < \infty$
- 4 By Uniform Boundedness Principle: $\sup_{x \in C} \|T_x\| < \infty$
- 5 This means C is bounded

Significance

This shows that weak boundedness (boundedness under all linear functionals) implies actual boundedness.

Finite Dimensionality Simplification

Theorem

In a finite-dimensional normed space X , **all linear operators** $A : X \rightarrow Y$ are:

- *Continuous*
- *Bounded*
- *Have closed graph*

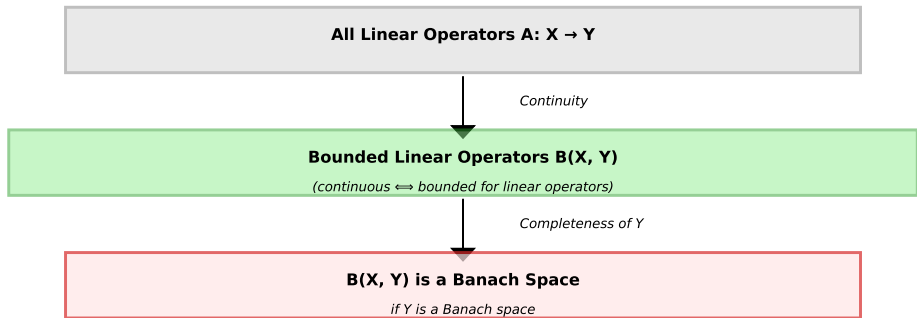
Why?

In finite dimensions:

- 1 All norms are equivalent
- 2 All linear operators can be represented as matrices
- 3 Matrices are continuous (polynomial function of entries)
- 4 No pathological unbounded operators exist

Summary of Key Results

Hierarchy of Operator Spaces



Key Properties of $B(X, Y)$:

- Norm: $\|A\|_{B(X,Y)} = \sup \{\|A(x)\|_Y : \|x\|_X \leq 1\}$
- Linear space under operator addition and scalar multiplication
- Normed space with complete metric if Y is Banach

Example 1: Computing Operator Norm

Problem

Let $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator represented by matrix:

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

Compute $\|A\|_B$ using the Euclidean norm.

Solution

The operator norm is the largest singular value of A :

- ① Compute $A^T A = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$
- ② Find eigenvalues: $\det(A^T A - \lambda I) = 0$
- ③ $(4 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 6\lambda + 4 = 0$
- ④ $\lambda = 3 \pm \sqrt{5}$
- ⑤ Largest eigenvalue: $\lambda_{\max} = 3 + \sqrt{5} \approx 5.236$

Example 2: Integral Operator on $L^2[0, 1]$

Problem

Let $X = L^2[0, 1]$ and define $A : L^2[0, 1] \rightarrow L^2[0, 1]$ by:

$$[Ax](t) = \int_0^1 K(t, s)x(s) ds$$

where the kernel is $K(t, s) = ts$. Is A bounded?

Solution

By Cauchy-Schwarz inequality:

$$\begin{aligned} |[Ax](t)|^2 &= \left| \int_0^1 ts \cdot x(s) ds \right|^2 \\ &\leq \int_0^1 t^2 s^2 ds \cdot \int_0^1 x(s)^2 ds \\ &= \frac{t^2}{3} \|x\|^2 \end{aligned}$$

Example 3: Checking Operator Continuity

```
import numpy as np
from scipy import linalg

# Define linear operator as a matrix
def operator_norm(A, norm_type='spectral'):
    """
    Compute operator norm of matrix A
    norm_type: 'spectral' (singular value),
               'frobenius', '1', 'infinity'
    """
    return linalg.norm(A, ord=norm_type)

# Example: A = [[2, 1], [0, 1]]
A = np.array([[2., 1.], [0., 1.]])

# Compute different norms
print("Spectral norm (largest singular value):",
      operator_norm(A, 'spectral'))
print("Frobenius norm:", operator_norm(A, 'frobenius'))
```


Operator Norm Definition

$$\|A\|_B = \inf \{M : \|A(x)\| \leq M\|x\|, \forall x \in X\}$$

Norm of bounded linear operator $A: X \rightarrow Y$

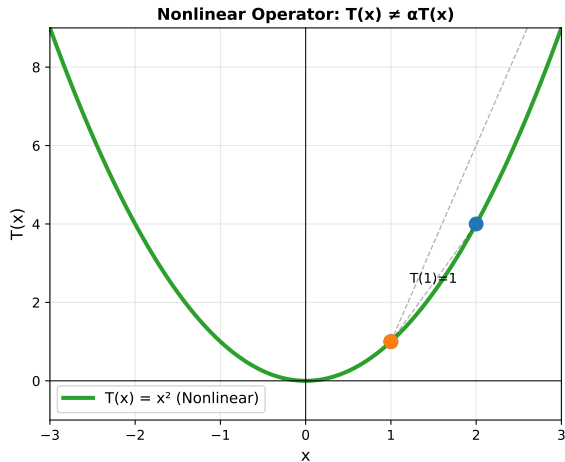
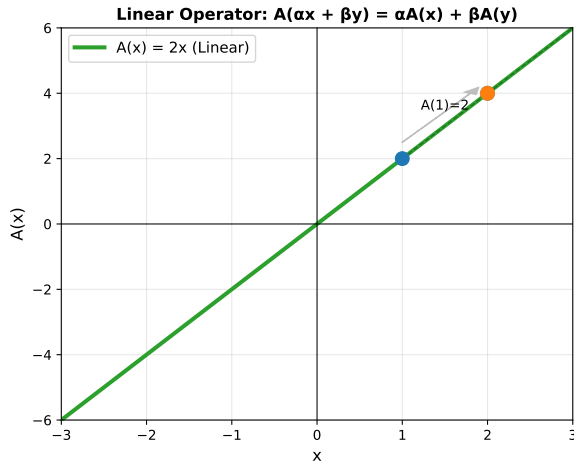
Equivalent forms:

- $\|A\|_B = \sup \{\|A(x)\| : \|x\| = 1\}$
- $\|A\|_B = \sup \{\|A(x)\|/\|x\| : x \neq 0\}$
- $\|A\|_B = \sup \{\|A(x)\| : \|x\| \leq 1\}$

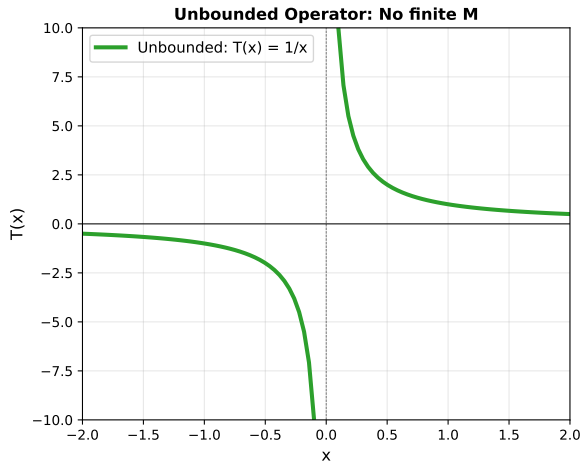
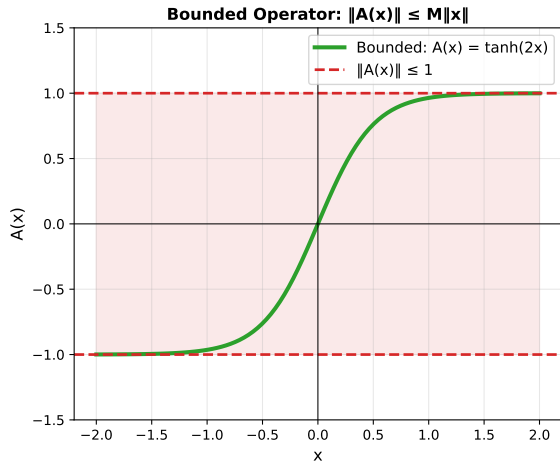
Properties of operator norm:

1. $\|A(x)\| \leq \|A\|_B \cdot \|x\|$ for all $x \in X$
2. $\|A + B\|_B \leq \|A\|_B + \|B\|_B$ (triangle inequality)

Visual Comparison: Linear vs Nonlinear

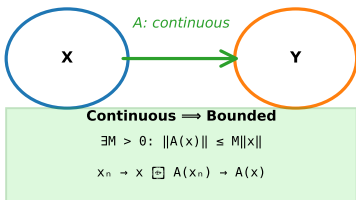


Bounded vs Unbounded Operators

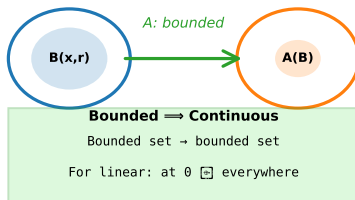


The Continuity-Boundedness Connection

Continuous Linear Operator



Bounded Linear Operator



Main Takeaways

- 1 **Linear operators** are the central objects in functional analysis, generalizing matrices to infinite-dimensional spaces
- 2 For linear operators: **boundedness** $\Leftrightarrow$ **continuity**
- 3 The **operator norm** $\|A\|_B$ uniquely determines boundedness and provides the standard for comparing operators
- 4 The space $B(X, Y)$ of bounded operators is itself a Banach space, enabling infinite-dimensional analysis
- 5 **Three types of convergence** exist: uniform (strongest), strong (medium), weak (weakest)
- 6 The **Uniform Boundedness Principle** connects pointwise and uniform boundedness—a corner stone result

Applications in Analysis and PDEs

Functional Analysis

- Spectral theory: eigenvalues and eigenvectors of bounded operators
- Riesz representation theorem: correspondence between functionals and elements
- Lax-Milgram theorem: solvability of variational problems

Numerical Analysis

- Iterative methods: convergence analysis via operator powers
- Finite element methods: approximation by bounded projection operators
- Error analysis: bounding errors using operator norms

PDEs and Physics

- Differential operators: Laplacian, divergence, curl
- Integral equations: Fredholm and Volterra operators
- Quantum mechanics: observables as self-adjoint operators

Further Reading and References

- **Pathak (2018)**. An Introduction to Nonlinear Analysis and Fixed Point Theory. Springer.
 - Chapter 1.3: Spaces of Bounded Linear Operators
 - Chapter 1.3.1: Properties of Bounded Linear Operators
 - Chapter 1.3.2: Convergence in $B(X, Y)$
- **Rudin (1991)**. Functional Analysis. McGraw-Hill.
 - Comprehensive treatment of operator theory
 - Spectral analysis and applications
- **Kreyszig (1978)**. Introductory Functional Analysis with Applications. Wiley.
 - Intuitive exposition with applications
 - Many worked examples and problems

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Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer-Verlag.



Rudin, W. (1991). *Functional Analysis* (2nd ed.). McGraw-Hill International.



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Conway, J. B. (1990). *A Course in Functional Analysis* (2nd ed.). Springer-Verlag.



Yosida, K. (1995). *Functional Analysis* (6th ed.). Springer-Verlag.



Brezis, H. (2010). *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Springer.

Linear Operators: The Bridge Between Linear Algebra and Functional Analysis

Key Insight

Bounded linear operators provide a natural and powerful framework for generalizing matrix theory to infinite dimensions, with profound applications throughout mathematics, physics, and engineering.

Questions? Discussion?